Statistical Learning and Korean L2 Learners' Production of English Relative Clauses

A number of studies have linked language learners' statistical learning ability – defined as the implicit ability to track the frequent patterns occurring in a language and to generalize rules -to their performance in language processing, suggesting that both are closely related (e.g., McDonough & Trofimovich, 2016; Misyak & Christiansen, 2012). Despite the findings that learners' statistical learning predicts their comprehension in L1 and L2 sentence processing, it remains unclear whether the statistical learning mechanism is associated with the language production. Extending these findings, the present study examined the hypothesis that learners' statistical learning ability, as measured in an artificial grammar learning task (e.g., Gómez [2002]), would be related to their performance in language production.

Twenty Korean learners of English (15 female and 5 male; mean age= 21.91, *SD* = 3.92, TOEIC average score = 851.6 out of 990) participated an artificial grammar learning (AGL) task and a picture-description task in one session. In the AGL task, the participants were exposed to nonadjacent dependency rules adapted from Gomez's (2002) study. They read four-word strings that followed the nonadjacent rules formed between the first position (*pel, dak, vot*) and the final position (*jic, rud, tood*), with twenty-four pairs of intervening nonwords (e.g., [1]). All grammatical sentences (n=144) were presented automatically one at the time. In the following test phase, participants judged the grammaticality of 40 sentences (20 grammatical and 20 ungrammatical; see [1] and [2]) based on what they had learned during the training.

For the English production task, the participants first heard a recorded sentence along with each picture (e.g., [3]), and, after the question (e.g., *Who do you see?*), described the object in the third picture (target answer: *a man who think of woman*). They produced16 items in total (8 subject relative clause and 8 object relative clause with four practice items. To examine the relation between statistical learning (SL) ability and language production, the accuracy and response time (RT: time from end of the question to the onset of the target) were measured in a picture-based production task. For the AGL task, the grammaticality accuracy and RTs were recorded.

In the AGL task, the participants' AGL accuracy was 70.48% (*SD*=25.35), significantly above chance level, and the AGL judgment RT was 742.09 ms (*SD*= 452.98). In the production task, the accuracy was 46.63% (SD= 38.66) for subject relative clauses and 43.75% (*SD*=36.8) for object relative clauses. The average production RT was 1.26 s (*SD* = 1.77) for subject relative clause and 0.98 s (*SD*= 1.44) for object relative clauses.

The main interest in this study was to examine the relation between learners' SL ability and production performance. Results showed that a significant correlation between AGL judgment RT and the response accuracy for subject relative clauses (r = 519, p < .05) and object relative clauses (r =.483, p <.05). The AGL judgment RT was also marginally related to the production RT for object relative clauses (r = - .421, p = .065). These findings indicated that learners' statistical learning is related to their ability to produce English relative clauses. A regression analysis further showed that SL ability might predict their production performance ((F (1,19) = 5.491, p = .031, R-squared = .234, adjusted R-squared = .191, β = .483, t =2.343, p =031), consistent with prior research showing its predictor in comprehending English relative clauses (e.g., Misyak & Christiansen, 2012).

A sample of materials used in an artificial grammar learning (AGL) task

Grammatical	Ungrammatical
[1] a. **pel** rob gik **jic** b. **dak** vot juf **rud** c. **vot** vev fuf **tood**	[2] a. * **pel** deg dap **tood** b. * **dak** bov bul **jic** c. * **vot** bup cag **rud**

[3] A sample of a material used in an English picture-based production task

 Slide 1 There is a woman.	 Slide 2 She broke up with her boyfriend.	 Slide 3 Question: Who do you see?
Targeting answer: a man who think of a woman		